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Lab Experiment 6

AIM

To understand the concept and working of cursors in PL/SQL and implement implicit and explicit cursors to process multiple rows and apply business logic.

Objective of the Session

- To learn how implicit cursors work in PL/SQL for single-row operations
- To implement explicit cursors for handling multiple records
- To use cursor attributes like %FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN
- To process records row-by-row and apply business logic

Software Requirements

Database Management System:

Oracle Database Express Edition (Oracle XE)

PostgreSQL Database

Database Administration Tool / Client Tool:

Oracle SQL Developer (for Oracle XE)

pgAdmin (for PostgreSQL)

<u>Experiment Question</u> :

Write PL/SQL programs to:

1. Use implicit cursors for single-row operations
2. Use explicit cursors to fetch and process multiple employee records
3. Use cursor attributes to control execution
4. Apply business logic (e.g., salary increment based on condition)

Practical/Experiment steps:

1. Create an **EMPLOYEE table**
1. Insert sample data
3.Write PL/SQL block using **implicit cursor**
4. Write PL/SQL block using **explicit cursor**
5. Use cursor attributes for control
6. Display results using DBMS_OUTPUT

CODE:

```
CREATE TABLE employee (
   emp_id INT PRIMARY KEY,
   emp_name TEXT,
   salary NUMERIC
);

INSERT INTO employee VALUES
(1, 'Manan', 25000),
(2, 'Rahul', 35000),
(3, 'Amit', 20000);


DO $$
DECLARE
   v_id INT;
   v_name TEXT;
```

```
    v_salary NUMERIC;

    emp_cursor CURSOR FOR
        SELECT emp_id, emp_name, salary FROM employee;

BEGIN
    -- Implicit-like update
    UPDATE employee
    SET salary = salary + 1000
    WHERE emp_id = 1;

    IF FOUND THEN
        RAISE NOTICE 'Row Updated';
        RAISE NOTICE 'Rows Affected: %', FOUND;
    END IF;

    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id, v_name, v_salary;
        EXIT WHEN NOT FOUND;

        IF v_salary < 30000 THEN
            UPDATE employee
            SET salary = salary + 5000
            WHERE emp_id = v_id;
        END IF;

        RAISE NOTICE 'ID: %, Name: %, Salary: %',
                    v_id, v_name, v_salary;

    END LOOP;

    CLOSE emp_cursor;

    RAISE NOTICE 'Processing Completed';

END $$;
```

OUTPUT:

```
Data Output   Messages   Notifications

NOTICE:  Row Updated
NOTICE:  Rows Affected: t
NOTICE:  ID: 2, Name: Rahul, Salary: 35000
NOTICE:  ID: 3, Name: Amit, Salary: 20000
NOTICE:  ID: 1, Name: simran, Salary: 26000
NOTICE:  Processing Completed
DO

Query returned successfully in 84 msec.
```

Learning Outcome

1. Understand how cursors help in processing multiple rows
2. Differentiate between implicit and explicit cursors
3. Use cursor attributes effectively
4. Implement row-by-row processing logic
5. Apply cursor-based solutions in real-world scenarios